

Likelihood Functions: Pt 2

CSU, Chico Math 351

2026-01-30

Poisson Distribution

Assume $x_1, \dots, x_n \sim_{iid} \text{Poisson}(\lambda)$, where λ is the rate of occurrences. Calculate the maximum likelihood estimator of λ with pen and paper or LaTeX, showing your work. The probability mass function (PMF) for the Poisson distribution is:

$$f(x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Poisson Distribution

With R, generate data from $Poisson(\lambda = 3)$ with `rpois`, then use `optim` to estimate λ via the log-likelihood function for the Poisson distribution.

Distributions with Multiple Parameters

Be warned that if a distribution has more than one parameter, `optim()` MUST read in as a single object! For instance, a distribution with the parameters $\theta = (a, b)$ would need to begin by using:

```
ll.function <- function(x, theta){  
  a <- theta[1]  
  b <- theta[2]  
}
```

The `optim()` function will not work as intended if you instead use "`function(x,a,b)`" since it takes in two arguments: data (`x`) and parameters (`theta`).

Weibull Distribution

With pen and paper or LaTeX, derive the maximum likelihood estimator for the **scale** parameter λ of the Weibull distribution, $Weibull(k, \lambda)$, given a known **shape** parameter $k = 3$. Show your work.

$$f(x|\lambda, k = 3) = \left(\frac{k}{\lambda}\right) \left(\frac{x}{\lambda}\right)^{k-1} \exp\left(-(x/\lambda)^k\right), x \geq 0.$$

Poisson Distribution

With R, generate data from $Weibull(k = 3, \lambda = 4)$ with `rpois`, then use `optim` to estimate λ via the log-likelihood function for the Poisson distribution.